

Lecture 2: The Simplex Method & Diameter of Polyhedra

Algorithms & Geometry of Linear Programming

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DIMAG Summer School on Combinatorics and Algorithms

August 2026

From geometry to algorithm

Lecture 1: a feasible, bounded LP is solved at a *vertex*, and local optimality implies global optimality.

Today: the **simplex method** [Dantzig '47] — walk from vertex to vertex along improving edges.

- **in practice**: number of pivots typically *linear* in the problem size [Shamir '87]
- **in theory**: exponential worst case: the **Klee–Minty cube** [Klee–Minty '72]

Second theme: how short can vertex-to-vertex paths be? — the **diameter** of polyhedra.

Today

The Landscape of LP Algorithms

The Simplex Method

Pivot Rules

The Diameter of Polyhedra

Outline

The Landscape of LP Algorithms

The Simplex Method

Pivot Rules

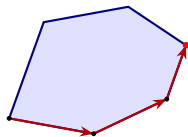
The Diameter of Polyhedra

Four families of LP algorithms

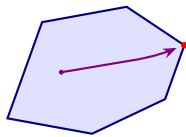
Two equivalent formulations; data $(\mathbf{A}, b, c)$, $\mathbf{A} \in \mathbb{R}^{m \times n}$; $L :=$ bit length:

$$\min c^T x \text{ s.t. } \mathbf{A}x = b, x \geq \mathbf{0} \quad (\text{standard})$$

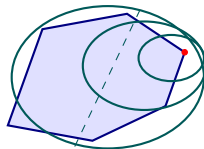
$$\max c^T x \text{ s.t. } \mathbf{A}x \leq b \quad (\text{geometric})$$



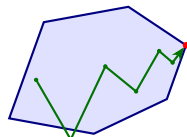
simplex
[Dantzig '47]



interior point
[Karmarkar '84]



ellipsoid
[Khachiyan '79]



first-order

Each iteration is a fixed amount of linear algebra $\Rightarrow$ compare **iteration counts**.

Exact versus approximate solvers

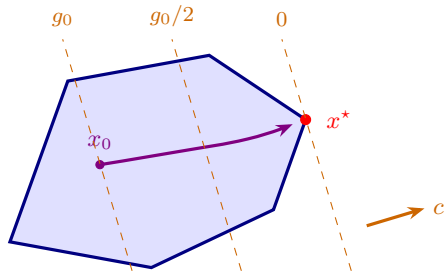
exact: terminate with an optimal vertex +
certificate (*simplex*)

*How many iterations to reach the
optimum?*

approximate: drive down the gap
 $g(x) := c^T x - \text{OPT}$
(*interior point, ellipsoid, first-order*)

How many iterations to halve the error?

each method reduces a computable **proxy**:
barrier parameter μ / ellipsoid volume /
KKT residual



halving the gap = advancing one dashed
line

From weakly to strongly polynomial

- over the rationals, approximate $\Rightarrow$ exact: gap below $2^{-\Theta(L)}$ $\Rightarrow$ round to the optimal vertex
- halving the gap every T iterations solves the LP in $O(T \cdot L)$ iterations: **weakly polynomial**

Definition (Strongly polynomial algorithm [Megiddo '83])

An exact LP algorithm is *strongly polynomial* if (i) it performs $\text{poly}(m, n)$ basic arithmetic operations, and (ii) intermediate numbers stay of bit length $\text{poly}(L)$.

- existence of one for LP: major open problem, on Smale's list for the 21st century [Smale '98]
- a $\text{poly}(m, n)$ pivot bound for *some* pivot rule would resolve it

Strongly polynomial algorithms: what is known

- maximum flow [Edmonds–Karp '72], minimum-cost flow [Tardos '85]
- $\mathbf{A}$ with polynomially bounded integer entries [Tardos '86] — running time independent of b and c
- interior point with iteration count depending only on $\mathbf{A}$ [Vavasis–Ye '96], via a condition measure $\approx$ the *circuit imbalance* $\kappa_{\mathbf{A}}$ of Lecture 3 [DHNV '20]
- ≤ 2 nonzeros per column of $\mathbf{A}$ [DKNOV '24]
- a limit: no central-path-following interior point method is strongly polynomial [ABGJ '18] — Lecture 4

Today: the exact side of the landscape — the simplex method.

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The vertex-edge graph, and the walk

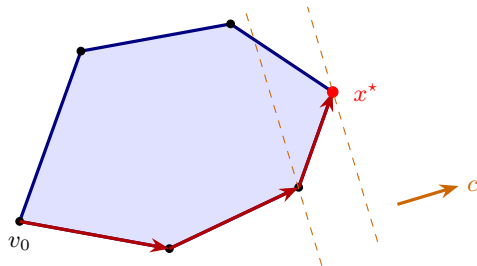
Throughout: $P = \{x : \mathbf{A}x \leq b\}$, $\text{rank}(\mathbf{A}) = n$, LP $\max\{c^T x : x \in P\}$.

vertex-edge graph $G(P)$: nodes = vertices,
edges = bounded 1-dim. faces

Abstract simplex method

While an incident edge *improves* the objective, move along it (ray $\Rightarrow$ *unbounded*). Output the final vertex.

choice of improving edge = the **pivot rule**



To justify: stuck $\Rightarrow$ optimal, and the walk terminates.

The local object: the tangent cone

Definition (Tangent cone)

For $x \in P$ with tight constraints $I(x) = \{i : a_i^\top x = b_i\}$:

$$T_P(x) := \{w : x + \varepsilon w \in P \text{ for some } \varepsilon > 0\} = \{w : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}.$$

seen from x , all of P lies in the cone:

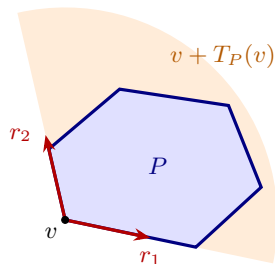
$$P \subseteq x + T_P(x)$$

Lemma (Edges generate the tangent cone)

At a vertex v : $T_P(v)$ is pointed and

$$T_P(v) = \text{cone}(e - v : e \text{ edge of } P \text{ at } v),$$

with extreme rays = the incident edge directions.



Correctness of the simplex method

Theorem

Let $v \in V(P)$. Exactly one of the following holds:

- (a) $c^T r \leq 0$ for every incident edge direction r : then v is *optimal*;
- (b) some incident edge direction has $c^T r > 0$: then moving along it reaches a strictly better vertex, or certifies that the LP is *unbounded*.

Proof. (a) For $y \in P$: $y - v \in T_P(v) = \text{cone}(\text{edge directions})$, say $y - v = \sum_r \mu_r r$, $\mu \geq 0$, so

$$c^T y - c^T v = \sum_r \mu_r c^T r \leq 0.$$

(b) The edge ends at a neighbor of strictly larger value, or is a ray $r \in \text{rec}(P)$ with $c^T r > 0$ — unbounded (Lecture 1). □

Termination — and degeneracy

Corollary (termination)

If every pivot strictly increases the objective, no vertex is revisited: at most $|V(P)| \leq \binom{m}{n}$ pivots, *for any pivot rule*.

- strict increase is guaranteed when no vertex has $> n$ tight constraints
- **degeneracy**: $> n$ tight constraints — a pivot may exchange tight constraints *without moving*; the pivot rule can cycle [Bland '77: anti-cycling]
- fix (exercise): perturb $b \mapsto b + (\varepsilon, \varepsilon^2, \dots, \varepsilon^m)^T$ — no vertex degenerate for small $\varepsilon > 0$, and an optimal basis of the perturbed LP is optimal for the original

We assume non-degeneracy today. The real question: **how many pivots?**

A byproduct: the vertex-edge graph is connected

Corollary

If P is pointed and non-empty, then $G(P)$ is connected.

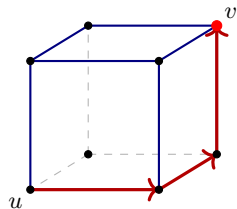
Proof. To reach a target vertex v from any u : build an objective maximized *uniquely* at v ,

$$c := \sum_{i \in I(v)} a_i \implies c^T y \leq \sum_{i \in I(v)} b_i = c^T v \quad \forall y \in P,$$

equality iff $y = v$ ($\text{rank } \mathbf{A}_{I(v)} = n$). Run simplex from u : bounded LP $\Rightarrow$ no rays taken; strict increase $\Rightarrow$ terminates — at the unique optimum v .

□

The walk itself traces the u – v path.



1-skeleton of the cube

The simplex method in standard form

Geometric form is the language of pictures; for *computation*, switch to

$$\min c^T x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0}, \quad \mathbf{A} \in \mathbb{R}^{m \times n} \text{ of full row rank.}$$

geometric form $\{x : \mathbf{A}x \leq b\}$	standard form $\{x : \mathbf{A}x = b, x \geq \mathbf{0}\}$
tight constraint $a_i^T x = b_i$	zero coordinate $x_j = 0$
vertex = n indep. tight constraints	BFS: $x_N = \mathbf{0}$, $\mathbf{A}_B$ invertible
edge = $n - 1$ indep. tight constraints	$x_{[n] \setminus S} = \mathbf{0}$, $\ker(\mathbf{A}_S)$ 1-dim. ($ S = m+1$)
tangent cone $\{w : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}$	$\{w \in \ker(\mathbf{A}) : w_j \geq 0 \text{ if } x_j = 0\}$
non-deg.: no vertex with $> n$ tight	non-deg.: $x_B > \mathbf{0}$ at every BFS

via *slack coordinates* $s := b - \mathbf{A}x$: constraint i tight $\iff s_i = 0$

Edge directions and reduced costs: algebra = geometry

Basis $B \subseteq [n]$, $|B| = m$, $\mathbf{A}_B$ invertible; BFS x : $x_B = \mathbf{A}_B^{-1}b \geq \mathbf{0}$, $x_N = \mathbf{0}$.

Non-degenerate ($x_B > \mathbf{0}$): tight constraints = $\{\mathbf{A}x = b\} \cup \{x_j = 0 : j \in N\}$.

Incident edge = relax *one* bound $x_j = 0$, keep the rest tight:

$$\mathbf{A}w^{(j)} = \mathbf{0}, \quad w_{N \setminus \{j\}}^{(j)} = \mathbf{0}, \quad w_j^{(j)} = 1 \quad \implies \quad w_B^{(j)} = -\mathbf{A}_B^{-1}\mathbf{A}e_j$$

objective slope along the edge, *per unit* of x_j :

$$c^\top w^{(j)} = c_j - c_B^\top \mathbf{A}_B^{-1} \mathbf{A}e_j =: \bar{c}_j \quad (\text{reduced cost; } \bar{c}_B = \mathbf{0})$$

$n - m$ directions, one per non-basic j = extreme rays of $\{w \in \ker(\mathbf{A}) : w_N \geq \mathbf{0}\}$.

Optimality test, fully explicit: $\bar{c} \geq \mathbf{0} \iff x$ optimal.

One iteration, in full

Simplex method, standard form, non-degenerate (one iteration)

Given: basis B with BFS x , $x_B = \mathbf{A}_B^{-1}b > \mathbf{0}$.

1. reduced costs $\bar{c}^\top = c^\top - c_B^\top \mathbf{A}_B^{-1} \mathbf{A}$; if $\bar{c} \geq \mathbf{0}$: x optimal — stop
2. *entering* index: choose j with $\bar{c}_j < 0$ (**the pivot rule**)
3. edge direction: $w_j = 1$, $w_B = -\mathbf{A}_B^{-1} \mathbf{A} e_j$, zero elsewhere
4. if $w_B \geq \mathbf{0}$: *unbounded* along the ray $x + t w$ — stop
5. ratio test: $t^* = \min_{i \in B: w_i < 0} x_i / (-w_i)$, attained at *leaving* index i^*
6. update $x \leftarrow x + t^* w$, $B \leftarrow B \cup \{j\} \setminus \{i^*\}$

cost per pivot: one linear system solve; non-degeneracy $\Rightarrow t^* > 0 \Rightarrow$ termination

Initialization: the two-phase scheme

The method needs a starting BFS — but finding *any* feasible point is a priori as hard as LP itself.

Resolution: feasibility is itself an LP *with an obvious starting vertex*.

Phase 0 (after flipping signs of rows so that $b \geq 0$)

$$\min \mathbf{1}^T z \text{ s.t. } \mathbf{A}x + z = b, \quad x, z \geq \mathbf{0}, \quad \text{starting BFS: } (x, z) = (\mathbf{0}, b)$$

- optimal value 0 $\iff$ the original LP is feasible
- an optimal BFS with $z = \mathbf{0}$ starts *Phase 1*: simplex on the real objective

Where we are

This hour:

- simplex method: walk along improving edges — correct for every pivot rule (tangent cone), terminating under non-degeneracy, $G(P)$ connected
- fully explicit in standard form: edges = kernel directions, slopes = reduced costs, one linear system per pivot; initialization via two phases

Wide open so far: how many pivots?

- does some clever choice of improving edge always take few steps?
- are there always *short* vertex-to-vertex paths at all?

This afternoon: pivot rules and the Klee–Minty cube; then the diameter of polyhedra.

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Classical pivot rules

A pivot rule decides which improving edge to take.

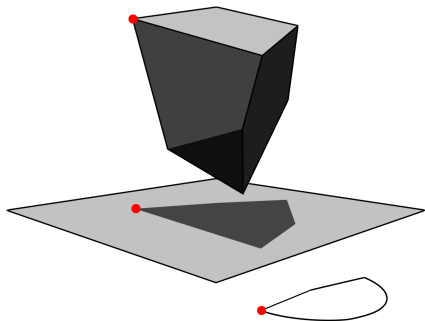
- **Dantzig's rule:** most negative reduced cost $\bar{c}_j$ — improvement *per unit* of the entering variable
- **greatest improvement:** maximize the actual decrease $t^*|\bar{c}_j|$ — needs a ratio test per candidate
- **steepest edge:** edge direction of smallest angle with $-c$ — empirically the strongest; its approximation *Devex* is the default in modern solvers

None is known to be efficient in the worst case; for most classical rules, explicit exponential examples exist.

The shadow vertex rule

Two objectives: c with *known unique minimizer* v_0 , and a *target* d .

1. start at the vertex v_0 minimizing c
2. slide the objective: $c_\lambda := (1 - \lambda)c + \lambda d$
3. increase λ from 0 to 1, tracking a c_λ -minimizing vertex

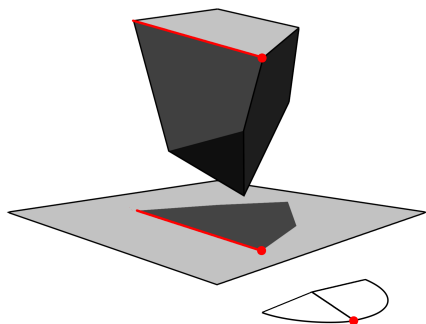


The walk (red) projects onto the boundary of the **shadow** of P on the plane $\text{span}(c, d)$. The wheel (bottom right) shows the *negated* objective $-c_\lambda$ turning from $-c$ to $-d$: the tracked vertex is the one *furthest* in that direction.

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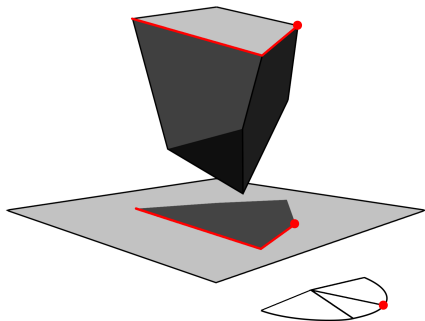


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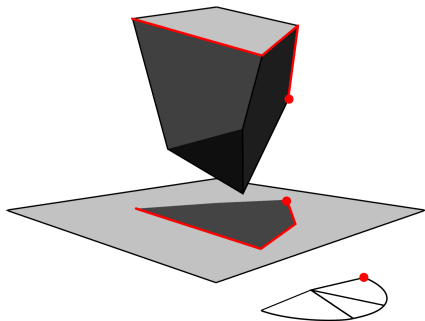


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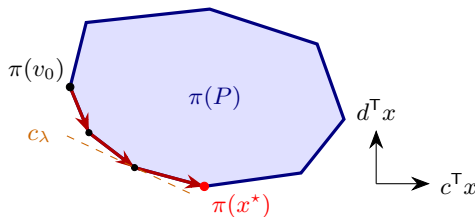
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The shadow vertex rule

Two objectives: c with *known unique minimizer* v_0 , and a *target* d . Slide:

$c_\lambda := (1 - \lambda)c + \lambda d$, $\lambda : 0 \rightarrow 1$, maintaining a c_λ -minimizing vertex.

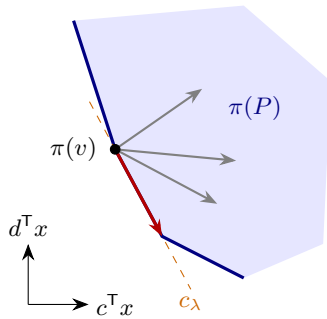
- breakpoints finitely many, consecutive optima adjacent: a simplex path $v_0 \rightarrow d$ -optimum
- vertex visited $\iff$ it minimizes *some* $c_\lambda \iff$ it projects onto the lower-left boundary of the shadow $\pi(P)$,
 $\pi(x) = (c^\top x, d^\top x)$
- that boundary = the **trade-off curve**:
best d -value for each c -value



$\# \text{pivots} \leq \# \text{vertices of } \pi(P)$

Shadow vertex: anchoring, implementation, locality

- **where does c come from?** any c with $c_B = \mathbf{0}$, $c_N > \mathbf{0}$ has unique minimizer v_0 — the tight-constraint trick again
- **which edge enters?** among edges progressing in the target ($d^\top w < 0$): minimize the **exchange rate** $c^\top w / (-d^\top w)$ — c -cost per unit of d -progress = steepest descent in the shadow plane
- **implementation:** $\bar{c}(\lambda)$ is *affine* in λ — a ratio test finds the next breakpoint
- **locality:** visited $\iff$ minimizes some c_λ — *not on the history of the walk*



edges point into $\pi(P)$: none is steeper than the boundary edge

Locality is what has enabled the probabilistic analyses of simplex — the trade-off curve returns in Lectures 3 and 4, always traversed from the c -optimal end.

The random facet rule

The current theoretically fastest method uses **randomization**. (Geometric form, dimension n , m facets.)

Random facet [Kalai '92]

At vertex v : if no improving edge, stop. Else pick a facet $F \ni v$ *uniformly at random*, recursively optimize over F (dimension $n - 1$). If F 's optimum is optimal for P , stop; else take an improving edge out of F and recurse.

- expected pivots $\leq e^{O(\sqrt{n \log m})}$ on every LP [Kalai '92; Matoušek–Sharir–Welzl '96] — **subexponential**
- still the state of the art; improved variant [Hansen–Zwick '15]; nearly matching lower bounds for the rule [Friedmann–Hansen–Zwick '11]

Exponential examples: the Klee–Minty cube

For a parameter $0 < \varepsilon < 1/2$ [Klee–Minty '72]:

$$\begin{array}{ll}\max & x_n \\ \text{s.t.} & 0 \leq x_1 \leq 1, \\ & \varepsilon x_{j-1} \leq x_j \leq 1 - \varepsilon x_{j-1}, \quad j = 2, \dots, n\end{array}$$

- pair j : a **floor** $\varepsilon x_{j-1} \leq x_j$ and a **ceiling** $x_j \leq 1 - \varepsilon x_{j-1}$ — a perturbed cube $[0, 1]^n$ with $2n$ facets
- floor and ceiling of a pair are never both tight — each vertex makes *exactly one* of each pair tight
- vertices $v^S \leftrightarrow$ subsets $S \subseteq [n]$ (pairs with tight ceiling), via forward substitution:

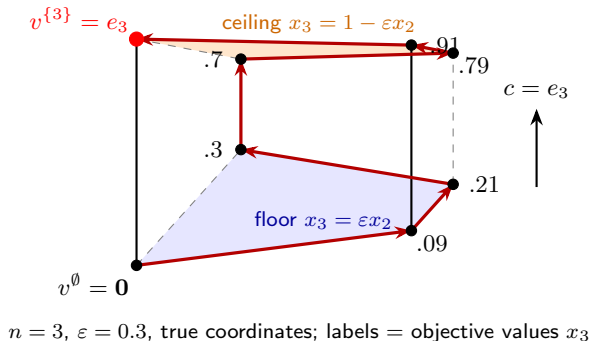
$$v_j^S = \begin{cases} 1 - \varepsilon v_{j-1}^S, & j \in S \\ \varepsilon v_{j-1}^S, & j \notin S \end{cases} \quad (v_0^S := 0)$$

A monotone Hamiltonian path

Theorem (Klee–Minty '72)

For $0 < \varepsilon < 1/2$, the graph of P_n has a Hamiltonian path from $v^\emptyset = \mathbf{0}$ to $v^{\{n\}} = e_n$ along which x_n strictly increases.

every step improves the objective,
yet *all* 2^n vertices are visited: an
adversarial improving pivot rule
takes $2^n - 1$ pivots, with only $2n$
facets

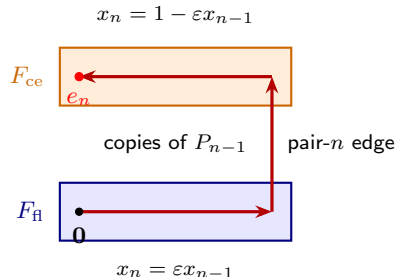


Proof idea

Induction on n :

- every vertex lies on exactly one pair- n facet:
floor $F_{\text{fl}} = \{x_n = \varepsilon x_{n-1}\}$ or ceiling
 $F_{\text{ce}} = \{x_n = 1 - \varepsilon x_{n-1}\}$
- both are *affine copies* of P_{n-1} :
 $y \mapsto (y, \varepsilon y_{n-1})$ and $y \mapsto (y, 1 - \varepsilon y_{n-1})$
- traverse the floor copy by induction
($x_n = \varepsilon x_{n-1}$ increases), cross the pair- n
edge, traverse the ceiling copy *in reverse*
($x_n = 1 - \varepsilon x_{n-1}$ increases as x_{n-1} decreases)

Start 0, end e_n . Full verification in the notes.



From adversarial paths to actual pivot rules

- the path defeats an *adversarially chosen* improving rule; forcing a *specific* rule requires calibrating the instance to the rule — on the instance as stated, Dantzig's rule reaches the optimum in *one* pivot!
- the calibration for **Dantzig's rule**: the recursive change of variables

$$x_j = \varepsilon x_{j-1} + \theta^{j-1} \tilde{x}_j \quad (0 < \theta < \varepsilon),$$

plus rescaling the ceilings — then Dantzig follows the Hamiltonian path exactly ($\varepsilon = \frac{1}{3}$, $\theta = \frac{1}{9}$: the classical formulation)

- similar constructions for virtually every deterministic rule pre-2000, including steepest edge and shadow vertex (*Goldfarb cube* — exercise)

Whether some pivot rule pivots in (strongly) polynomial time: **central open question**.

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The Simplex Method

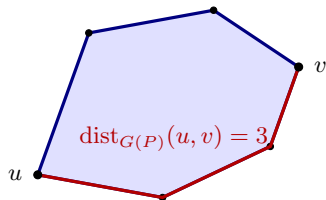
Pivot Rules

The Diameter of Polyhedra

A lower bound against every pivot rule

Back to geometric form: $P = \{x : \mathbf{A}x \leq b\}$, m facets; tight set $I(v) := \{i : a_i^\top v = b_i\}$.

- moving between vertices u, v takes $\geq \text{dist}_{G(P)}(u, v)$ pivots — *for every pivot rule*
- **combinatorial diameter:**
 $\text{diam}(P) := \max_{u, v \in V(P)} \text{dist}_{G(P)}(u, v)$
- a family with super-polynomial diameter would defeat every pivot rule — none is known



Polynomial Hirsch conjecture (open)

Is $\text{diam}(P) \leq \text{poly}(m, n)$ for every n -dimensional polyhedron P with m facets?

The Hirsch conjecture, and what is known

- Hirsch, 1957 (letter to Dantzig): is $\text{diam}(P) \leq m - n$?
- for polytopes: stood for over fifty years, until a counterexample [Santos '12]
- all known lower bounds remain within a *constant factor* of $m - n$

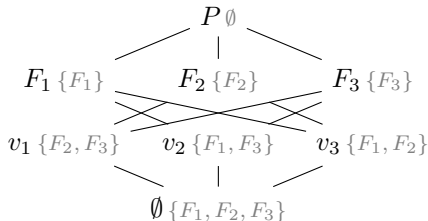
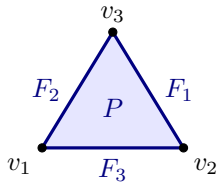
Theorem (Kalai–Kleitman '92)

Every n -dimensional polytope P with m facets satisfies $\text{diam}(P) \leq m^{1+\log_2 n}$.

Goal of this section: prove it — using almost no geometry.

Vertices as sets of facets: the face lattice

The faces of P ordered by inclusion: the **face lattice** — all the combinatorics, none of the metric data. Each face $\leftrightarrow$ its set of tight constraints (inclusion *reversing*).



- isomorphic face lattices = *combinatorially equivalent*: e.g. Klee–Minty $P_n \cong [0, 1]^n$

Simple polytopes

Assume P **simple**: every vertex on exactly n facets. (WLOG for diameter upper bounds: the perturbation $b + (\varepsilon, \dots, \varepsilon^m)^T$ from this morning makes P simple, and can only increase the diameter.)

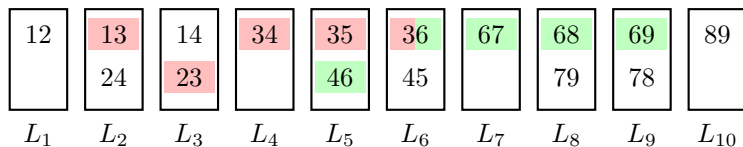
- $v \mapsto I(v)$ is injective into $\binom{[m]}{n}$
- adjacency is set intersection: $uv \in E(P) \iff |I(u) \cap I(v)| = n - 1$
- faces containing $v \leftrightarrow$ subsets of $I(v)$: $J \subseteq I(v)$ gives a face of dimension $n - |J|$

The diameter is determined by the set family $\{I(v) : v \in V(P)\}$ alone — an invitation to **discard the geometry**.

Connected layer families

Definition (CLF)

A *connected layer family* of order d on N symbols: a sequence $L_1, \dots, L_\ell$ of nonempty collections of d -element sets, all sets distinct, such that: for all $i \leq j \leq k$ and $A \in L_i$, $C \in L_k$, some $B \in L_j$ has $B \supseteq A \cap C$. The length is ℓ .



Each symbol occupies a **consecutive interval** of layers: symbol 3 (**red**) in layers 2–6, symbol 6 (**green**) in 5–9.

CLF calculus

Two closure properties drive everything (proofs: exercise):

Lemma (CLF calculus)

Let $L_1, \dots, L_\ell$ be a CLF of order d on N symbols.

- (i) *Sublayers: any subsequence $L_{i_1}, \dots, L_{i_r}$ is a CLF of order d .*
 - (ii) *Restriction: for a set S of $\leq d - 1$ symbols, the layers containing a set $\supseteq S$ form a contiguous interval; restricting them to those sets and deleting S gives a CLF of order $d - |S|$ on $\leq N - |S|$ symbols, of the same length.*
- (ii) with $S = \{s\}$: the interval property of the previous slide.

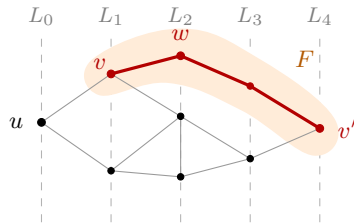
Polytopes yield CLFs

Lemma

P simple with m facets, $u \in V(P)$: the distance layers

$L_i := \{I(v) : \text{dist}_{G(P)}(u, v) = i\}$ form a CLF of order n on m symbols, of length $1 + \max_v \text{dist}(u, v)$.

- for $A = I(v)$, $C = I(v')$: the face F where $A \cap C$ is tight has a **connected** graph — the morning's corollary
- a $v-v'$ path in F crosses every layer in between: at the layer- j crossing, $I(w) \supseteq A \cap C$



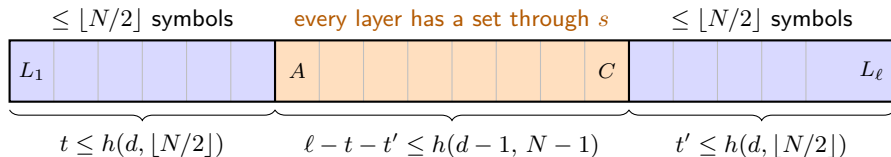
the path in F supplies a witness in every layer

Choosing u in a diameter pair: $\text{diam}(P) \leq \max \text{CLF length} - 1$.

The Kalai–Kleitman recursion

$h(d, N) :=$ maximum length of a CLF of order d on N symbols (0 if none exists, e.g. $N < d$).

$$h(d, N) \leq 2h(d, \lfloor N/2 \rfloor) + h(d-1, N-1) \quad (d \geq 2)$$



longest prefix/suffix on $\leq \lfloor N/2 \rfloor$ symbols; extended by one layer, each uses $> \lfloor N/2 \rfloor$ symbols $\Rightarrow$ they **share a symbol s** $\Rightarrow$ restrict & delete s : order drops to $d-1$.

Solving the recursion

Theorem (Abstract Kalai–Kleitman)

$$h(d, N) \leq N^{1+\log_2 d} = (2d)^{\log_2 N} \quad \text{for } d \geq 1, N \geq 2. \quad (\text{Hence } \text{diam}(P) \leq m^{1+\log_2 n}.)$$

Induction on N , all orders at once ($d = 1$: singletons, $h \leq N$; $N = 2$: $h \leq 2$). Unroll the recursion in its last term until the order reaches 1:

$$h(d, N) \leq 2 \sum_{i=2}^d h(i, \lfloor N/2 \rfloor) + N \leq 2(d-1) (2d)^{\log_2 N-1} + 2 \cdot (2d)^{\log_2 N-1} = (2d)^{\log_2 N}.$$

- each summand: induction at $\lfloor N/2 \rfloor$ gives $h(i, \lfloor N/2 \rfloor) \leq (2i)^{\log_2(N/2)} \leq (2d)^{\log_2 N-1}$; and $N = 2 \cdot 2^{\log_2 N-1} \leq 2(2d)^{\log_2 N-1}$
- refined bookkeeping: $\text{diam}(P) \leq (m - n)^{\log_2 n}$ [Todd '14] — still quasi-polynomial

Barnette–Larman: linear in m , exponential in n

Kalai–Kleitman halves the number of *symbols*; an older argument reduces the *order* instead.

Theorem (Barnette '74, Larman '70)

Every CLF of order d on N symbols has length $\leq 2^{d-1}N$. Consequently $\text{diam}(P) \leq 2^{n-1}m$.

- far better in fixed dimension: *linear* in the number of facets
- proof idea: greedily partition the layers into symbol intervals; restriction drops the order to $d - 1$ on each; each symbol meets ≤ 2 consecutive intervals
- guided proof in the exercises

How long can a connected layer family be?

The abstraction retains almost no geometry — yet **no super-polynomial CLF is known**.

- longest known: length $\Omega(N^2 / \log N)$, with order $d = \Theta(N)$ [EHRR '10]

Hähnle's conjecture (open; posed during Polymath 3: 2010-2011)

Every CLF of order d on N symbols has length at most $d(N - 1) + 1$.

- known in extremal cases; attained for *multiset* layers; even $d = 3$ is open ($4N - 1$ vs. conjectured $3N - 2$)
- if true: $\text{diam}(P) \leq n(m - 1)$ — far stronger than polynomial Hirsch

Summary and what comes next

Today:

- simplex method: correct for every pivot rule (tangent cone); explicit in standard form — edges = kernel directions, slopes = reduced costs
- pivot rules: Dantzig, steepest edge, shadow vertex (trade-off curve!), random facet ($e^{O(\sqrt{n \log m})}$, best known); Klee–Minty forces $2^n - 1$ improving pivots
- the diameter lower-bounds every pivot rule; Hirsch's $m - n$ fails [Santos '12]; polynomial Hirsch wide open
- Kalai–Kleitman: $\text{diam}(P) \leq m^{1+\log_2 n}$, purely combinatorially (CLFs)

Tomorrow: why is simplex fast *in practice*? **Smoothed analysis:** worst-case instances do not survive small random perturbations of the data [Spielman–Teng '04].

Exercises: removing degeneracy by perturbation, the Goldfarb cube, CLF calculus, Barnette–Larman.